

G shock — baseline transition: aggregates and GBC

Source: `fiscal_results.json` (G shock, baseline block). Figures checked: `g_macro_overview.png`, `g_fiscal_decomp.png`, `g_prices_sanity.png`, `g_debt_fan_chart.png`. $T_{\text{balance}} = 60$; 80 periods.

Main aggregates

- Y: flat, $0.449 \rightarrow 0.452$ (+0.6%); in the SOE at fixed r , K/L and Y/L are pinned, so output moves only with L .
- K_{domestic} : flat, $1.646 \rightarrow 1.657$; pinned to L by the firm FOC at the fixed world r .
- L : flat, $0.237 \rightarrow 0.238$.
- C : rises, $0.269 \rightarrow 0.273$ (+1.4%).
- w : constant at 1.271; exogenous in the SOE.
- r : constant at 0.040; the exogenous world rate.
- A (household wealth): rises, $1.794 \rightarrow 1.823$ (+1.6%).
- A/Y : rises slightly, $3.996 \rightarrow 4.034$.
- NFA/Y : falls steadily, $-1.32 \rightarrow -4.26$; with A/Y and K/Y roughly flat, NFA/Y declines nearly one-for-one with the rising B/Y ($NFA = A - K_{\text{domestic}} - B$).
- B/Y : rises $\times 2.8$, $1.646 \rightarrow 4.626$; r (4%) exceeds growth (≈ 0) and the $\approx 2\%$ -of- Y primary surplus is below the debt-stabilizing level.
- K_g/Y : not present in this run (no public capital; panel empty).

Government budget constraint (shares of Y)

- total_revenue : flat, $0.344 \rightarrow 0.346$.
- tax_l : flat, 0.070.
- tax_c : flat, $0.109 \rightarrow 0.110$.
- tax_p : constant, 0.130.
- tax_k : flat, 0.036.
- pension : rises slightly, $0.159 \rightarrow 0.162$ (population aging).
- ui : flat, 0.013.
- gov_health : flat, 0.054.
- govt_spending (G) : constant, 0.130 (pinned as a share of Y).
- $\text{public_investment (I}_g)$: constant, 0.030.
- defense_spending : constant, 0.030.
- $\text{other_net_spending}$: constant, -0.091 ; calibration residual pinned at the initial steady state.
- bequest_transfers : flat, 0.039.
- $\text{interest_payments (r} \cdot B)$: rises $\times 2.8$, $0.066 \rightarrow 0.185$; tracks the rising B/Y at $r = 4\%$.
- $\text{primary surplus (= -primary_deficit/Y)}$: flat, $0.0205 \rightarrow 0.0186$; held near the calibration target $\text{primary_balance_target_over_Y} = 0.0195$.

G shock — debt-financed vs baseline

Policy: G raised by 2% of Y from $t=0$, financed entirely by debt (no tax change); `balance_condition = terminal_debt_gdp`; no instrument adjustment; converged. Real allocation is unchanged because at the fixed world τ there is no crowding out and G does not enter the household problem.

Main aggregates vs baseline

- Y: unchanged ($\Delta = 0$ at every period).
- K_{domestic} : unchanged ($\Delta = 0$); K/L is pinned by τ , which does not move.
- L: unchanged ($\Delta = 0$).
- C: unchanged ($\Delta = 0$); households face the same prices, taxes, and lifetime budget.
- A: unchanged ($\Delta = 0$).
- A/Y : unchanged ($\Delta = 0$).
- NFA/Y : far below baseline, -2.29 at T_{balance} and -3.96 at the end (in units of Y); the extra debt is absorbed abroad one-for-one (A and K_{domestic} unchanged).
- B/Y : far above baseline, $+2.29$ at T_{balance} and $+3.96$ at the end (8.58 vs 4.63), as the unfunded G accumulates.
- w, r : unchanged (exogenous).

Government budget constraint vs baseline

- `govt_spending` (G): $+2.0$ pp of Y at all periods (the shock).
- `interest_payments` (rB): above baseline and widening, $+9.2$ pp of Y at T_{balance} and $+15.8$ pp at the end, compounding on the higher debt.
- primary surplus: 2.0 pp of Y below baseline at every period, a constant gap (G up 2 pp, revenue unchanged); because the baseline surplus is only $\approx 1.9\%$ of Y, the debt-financed level itself runs from $+0.05\%$ at $t=0$ to -0.14% (a small primary deficit) at the end.
- `total_revenue`, `tax_l`, `tax_c`, `tax_p`, `tax_k`, `pension`, `ui`, `gov_health`, `public_investment`, `defense_spending`, `other_net_spending`, `bequest_transfers`: unchanged from baseline ($\Delta = 0$).

G shock — labour-tax financed, debt target vs baseline

Policy: G raised by 2% of Y, financed by labour tax; `balance_condition = terminal_debt_gdp` pinned to the baseline terminal B/Y; $\Delta\tau_l = +3.17$ pp; converged. Higher τ_l lowers lifetime income: households work more, consume less, and save less.

Main aggregates vs baseline

- Y: above baseline, +0.0045 (+1.0%) at `T_balance` and the end; the income effect on labour raises L.
- `K_domestic`: above baseline, +0.017 (+1.0%); rises with L at fixed K/L.
- L: above baseline, +0.0024 (+1.0%); income effect of the tax dominates.
- C: below baseline, -0.0076 (-2.8%); lower after-tax lifetime income.
- A: below baseline, -0.041 (-2.2%); reduced saving.
- A/Y: below baseline, -0.13 of Y throughout.
- NFA/Y: below baseline, -0.14 at `T_balance` and -0.17 at the end (in units of Y); tracks the lower A/Y (K/Y unchanged, B/Y near baseline).
- B/Y: at baseline by construction at `T_balance` (+0.014), drifting +0.04 above by the end.
- w, r: unchanged (exogenous).

Government budget constraint vs baseline

- `tax_l`: +2.2 pp of Y (the +3.17 pp rate rise on the labour base).
- `total_revenue`: +1.7 pp of Y, from the higher labour tax.
- `govt_spending` (G): +2.0 pp of Y (the shock).
- primary surplus: -0.1 pp of Y; the tax rise nearly offsets the higher G, leaving B/Y on its target path.
- `interest_payments` ($r \cdot B$): $\approx$ baseline (+0.06 pp at `T_balance`, +0.17 pp at the end), since B/Y is held to target.
- pension: -0.16 pp of Y; higher Y lowers the pension share.
- `public_investment`, `defense_spending`, `other_net_spending`, `bequest_transfers`, `tax_c`, `tax_p`, `tax_k`, `ui`, `gov_health`: $\approx$ unchanged from baseline.

G shock — labour-tax financed, NFA@T target vs baseline

Policy: G raised by 2% of Y, financed by labour tax; `balance_condition = terminal_nfa_gdp` pinned to the baseline terminal NFA/Y; $\Delta\tau_l = +3.47$ pp; converged (to the closure's bracket, gap +0.018 of Y). The larger tax rise consolidates the budget enough to return external assets to baseline at T.

Main aggregates vs baseline

- Y: above baseline, +0.0049 (+1.1%); same income-effect channel as the debt-target case, slightly stronger.
- K_{domestic}: above baseline, +0.018 (+1.1%).
- L: above baseline, +0.0026 (+1.1%).
- C: below baseline, -0.0086 (-3.1%); the largest consumption drop of the three closures.
- A: below baseline, -0.052 (-2.9%).
- A/Y: below baseline, -0.16 of Y throughout.
- NFA/Y: returns to baseline at T_{balance} (+0.018 of Y, the target) and runs +0.12 of Y above by the end.
- B/Y: below baseline, -0.18 at T_{balance} and -0.28 at the end; the extra consolidation lowers debt to offset the lower A/Y and pin NFA/Y.
- w, r: unchanged (exogenous).

Government budget constraint vs baseline

- τ_l : +2.4 pp of Y (the +3.47 pp rate rise).
- total_revenue: +1.8 pp of Y.
- govt_spending (G): +2.0 pp of Y (the shock).
- primary surplus: +0.05 pp of Y; runs slightly tighter than baseline to push B/Y down.
- interest_payments (r·B): below baseline, -0.70 pp at T_{balance} and -1.12 pp at the end, on the lower debt stock.
- pension: -0.17 pp of Y; higher Y lowers the pension share.
- public_investment, defense_spending, other_net_spending, bequest_transfers, τ_c , τ_p , τ_k , ui, gov_health: $\approx$ unchanged from baseline.

Figure cross-check

Consistent. `g_fiscal_decomp.png` plots the primary deficit, the four taxes, UI, and pensions as shares of Y and the remaining lines (G, I_g, defense, other-net, interest) as levels; Y is $\approx$ flat so level and share directions coincide. The debt-financed B/Y explosion and its NFA/Y mirror, the τ_l scenarios' lower A/Y and C, and the NFA@T scenario's NFA/Y returning to baseline at T all match their macro-overview panels (B/Y also in the fan chart); w and r are flat and identical across scenarios in the prices panel; the K_g/Y panel is empty, matching the absent K_g.